

One percentage from another



1 Find the original number if:

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|-----------------------------------|-----------------------------------|
| a 1% of the number is 20. | b 2% of the number is 8. |
| c 10% of the number is 5. | d 20% of the number is 10. |
| e 25% of the number is 13. | f 50% of the number is 14. |
| g 5% of the number is 3. | h 10% of the number is 12. |

2 For each of the following:

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|-------------------------------------|---|
| i Find the 1% of the number. | ii Hence, find the whole number. |
| a 9% of the number is 72. | b 9% of a number is 63. |
| c 60% of a number is 480. | d 47% of a number is 188. |
| e 4% of a number is 236. | f 15% of a number is 7500. |
| g 12% of a number is 576. | h 8% of a number is 230. |

3 For each of the following:

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| i Find the 1% of the quantity. | ii Hence, find the whole quantity. |
| a 5% of a quantity is \$320. | b 11% of a quantity is \$16.50. |
| c 30% of a quantity is 2700 mL. | d 54% of a quantity is 6210 mL. |
| e 14% of a quantity is 7280 g. | f 22% of a quantity is 88 000 g. |
| g 34% of a quantity is 918 L. | h 78% of a quantity is 10 959 cm. |

4 For each of the following:

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|--------------------------------------|---|
| i Find 10% of the quantity. | ii Hence, find the whole quantity. |
| a 150% of a quantity is 135. | b 240% of a quantity is 1464. |
| c 80% of a quantity is \$160. | d 320% of a quantity is \$13 280. |

5 For each of the following:

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| i Find the 25% of the quantity. | ii Hence, find the whole quantity. |
| a 125% of a quantity is 30. | b 50% of a quantity is 85. |
| c 150% of a quantity is 75 000. | d 250% of a quantity is 1500. |

6 For each of the following, find the whole quantity:



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|----------------------------------|----------------------------------|
| a 14% of a quantity is 637. | b 26% of a quantity is 10 400. |
| c 32% of a quantity is \$10 240. | d 120% of a quantity is \$87.30. |
| e 80% of a quantity is 840 mL. | f 57% of a quantity is 4651.2 g. |
| g 64% of a quantity is \$10 400. | h 92% of a quantity is 4738 m. |

7 If 36% of a quantity is equal to 2412:

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| a Find 1% of the quantity. | b Find 83% of the quantity. |
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8 If 6% of a number is equal to 36:

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| a Find 1% of the number. | b Find 36% of the number. |
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9 If 21% of a quantity is equal to 378:

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| a Find 1% of the quantity. | b Find 71% of the quantity. |
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10 870% of a number is 696. Find the original number.

11 If 48% of 85 is 40.8, find 85% of 48.

Applications

12 $33\frac{1}{3}\%$ of a quantity is 448 grams. Find $66\frac{2}{3}\%$ of the same quantity.

13 462 students represent 30% of the school population. Find the number of students that make up 60% of the school population.

14 A city has a yearly water supply of 27 000 000 megalitres. It gets 14% of this water supply from melted snow. Find the amount of water that comes from melted snow each year.

15 When a rock sample was examined, it was found that 4.1 kilograms of it was copper. If this sample represents 10% of the total copper in the rock bed, how many kilograms of copper were there in the rock bed?

16 The 64 girls at Victoria's party represent 80% of all the guests. Find the total number of guests at the party.



- 17** Beth spends 90% of her weekly income on rent. If her rent is \$720 each week, find her weekly income.
- 18** After 10% GST was added, the price of an iPod was \$238. Calculate the original price before GST.
- 19** A total of 36 000 engineers were hired last year. This represents 9% of the total number of engineers employed in the country. How many engineers are employed in the country altogether?
- 20** Iain spent some time volunteering in a community. He spent 10% of his time volunteering at schools and 2% of his time volunteering at the local hospital. The remaining 132 days he spent volunteering at animal shelters.
How many days did Iain volunteer at the local hospital?
- 21** Oprah has completed 50% of the necessary 60 hours of pilot training.
- a** Find 50% of 60 hours.
 - b** Ryan has completed 5% more of the required training hours than Oprah. How many hours has he completed?
- 22** Fred's new cable television bill is \$98 per month. This is 140% of his monthly bill from last year, when he had the basic cable package. What was his monthly cable television bill last year?
- 23** Gwen has \$200 to spend on a new laptop. The sale price of the laptop includes the original price plus a sales tax of 9%. Find the greatest original price of the laptop that Gwen can afford. Round your answer to the nearest dollar.